Book File Format

On-disk records, properties, and geometry

File Header

Each .tbk begins with a 0xC00-byte PE stub containing no executable code. The real book starts at the ZZZTBKBOOKLOC marker (file offset 0xC00 + magic 0x22222222). Book signature is typically 03 4A 42 4F 01 00 00 08 (JBO + version).

Page Records

Page objects are the only records that begin with the marker **80 65 00 05 00 02 02**. Name is a 32-byte field. Property block at +43 starts 14 00 00 01. Widgets are nested 14 00 00 chunks inside the page.

Property IDs (from TB80BAS resource)

ID	Name	Notes
0x4004	bounds	Object bounds (twips)
0x402e	text	Field text (length-prefixed, often leading CRLF)
0x4023	script	OpenScript bytecode
0x4034	vertices	Point array route
0x4050	point-array setter	Runtime only; stores count + points
0x4015	idNumber	Object ID

Polygon Geometry (solved)

Draw records are 14 00 00 subtype-0 chunks with +0x0c = 04 00. Point count is the word at +0x1a; x,y twip pairs start at +0x22. Fill color dword is at +4 (e.g. 0xC0C0C0 silver). Screen coordinates = (stored * 96 + 720) / 1440 (or stored/15 when DPI factors are 96). Confirmed on P430 pin tips and chip body. Individual pin positions use an index at +0x52.

Arc Geometry

Type 0x11. Four twip pairs at +0x25/+0x27, +0x29/+0x2B, +0x2D/+0x2F, +0x31/+0x33. Bounding box at +0x0F. Fill color at +0x18. Painter is TB80BAS segment 0x7B. ToolBook does not call GDI!PIE; pie wedges are computed via GDIELLIPSEPOINTFROMANGLE then filled.

Object Type Table

TB80UTL.DLL at file 0xDED6 contains the classification table: ComboBox, Stage, Picture, Hotword, RecordField, PaintObject, Curve, AngledLine, Pie, Arc, IrregularPolygon, Polygon, Line, RoundedRectangle, Ellipse, Group, Field, Button, Rectangle, then Page, Background, Book. Resource kinds after *MEMBLOCK* include cel, anim, ref, sharedscript, clip, font, menubar, palette, cursor, icon, bitmap.

Open Items

- Absolute bounds are not stored; position is relative to a parent.
- Many connecting lines and detail strokes are computed from object bounds by segment 0x5B rather than

stored.
• Curve type ID is 0x05C7; control structure for embedded point groups still incomplete.
• Record-to-bounds extractor (the code that reads a 14 00 00 chunk and writes object+0xF) has not been located.